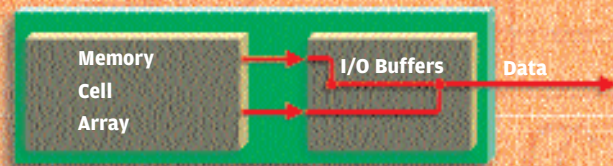


this component one of the most frequently upgraded.

To represent bandwidth in a simple formula: $\text{Bandwidth} = \text{Frequency} \times \text{Bus Width}$.

"Frequency" here means the rate of data transfer, (mostly measured in MHz), while bus width is measured in bits and represents the width of the path that the data flows through. Try to visualise RAM as a highway with multiple lanes. Traffic could be regarded as data. The wider the lanes, that is, the bus width, the more data can pass through in a given amount of time. Conversely, greater data speed (the speed of the traffic) will give the same result—more throughput. The first assumption about increasing the bus width of the memory isn't practical for all purposes. Memory width remains at 64 bits (remember it's DDR, so that's actually 32×2).

DDR 1 Memory Multiplexing



DDR Data flow diagram: note the 2-level multiplexing at work, represented by the two red data lines flowing into the I/O Buffer

Power consumption-wise, DDR2 is the hands-down winner, with a 1.8 V operating voltage as compared to the 2.5 V that DDR requires. The consumption of power will only increase as density (in gigabytes) of memory modules increases. For example, 4 GB of DDR memory would consume close to 40 watts of power for every read operation. Keeping in mind the universal drive for power gains through saving, DDR2 keeps environmentalists happy as well!

Another important advantage is DDR2's for the taking: due to its lower power consumption, much higher-density modules will be possible, and DDR2 memory can be available in 2 and 4 GB densities on a single DIMM! This is also possible due to the fact that DDR2 chips use an FBGA (Fine Ball Grid Array) packaging, while DDR utilises the older, albeit cheaper, TSOP (Thin Small-Outline Package) packaging, which was a serious hindrance to speed increments due to the inherent high resistance and inductance capacity in TSOP.

Architectures: DDR And DDR2 Compared

Let's look at a simple illustration showing data flow in DDR memory:

As we can see, there's a multiplexing of sorts taking place, since each bit from both pipelines is being fed onto the same output line. Here, "multiplexing" quite simply means sending multiple streams of data (let's look at it as a single bit of data for simplicity), at the same time and on the same stream, which are collated in the Input/Output buffer. We therefore have two bits at the end of one clock cycle for DDR memory. What happens is that both bits of data are captured at separate stages, fed into the pipeline, and then moved to the data bus, one each on the peak and trough clock signals. Each of the pipelines are 32 bits wide (there are two of them); the memory bus width is therefore 64 bits.

DDR2 RAM follows the double-up principle, by doubling the number of bits fetched from the data pipeline. The pipeline has been streamlined further to a 16-bit architecture, and the multiplexing principle is taken to the next level with four bits being supplied at each peak and trough. Therefore the 64 bits on a module are comprised of simultaneous transmission from each of the four banks.

Let's look at this in more depth. There are three different frequency values we need to

For example, DDR 400 MHz memory runs at 200 MHz internally (200×2 because it's DDR), while the bit width is 64, which gives us a bandwidth figure of 3200 MBps. In megabits, we'd get $400 \times 64 = 25600$ megabits per second. To arrive at the figure in megabytes, $25600/8 = 3200$ (MBps or megabytes per second). DDR 400 MHz memory is therefore also called PC 3200 RAM.

If we look at the bandwidth figure of a 32-bit, 3.2 GHz processor:

$3200 \times 32 = 102400$ MBps, or 12800 MBps.

Add to this the fact that memory has a latency—a wasted-clock-cycles figure that is astronomical as compared to that of a processor—and it's no wonder that even the fastest memory available bottlenecks today's CPUs.

DDR peaks out at 500 MHz, with some manufacturers going into the 550 MHz realm. But the problem was, DDR architecture was left with no headroom for future speed-ups. Although almost no amount of bandwidth increments will bring memory anywhere close to satisfying the processor's bandwidth requirements, DDR2 seems like a step in the logical direction, offering steep increments in raw clock speeds at least. With DDR2 promising that 1066 MHz will become mainstream by late 2007, and 800 MHz already available in goodly doses, it seems right on track to greet the latest monsters from Intel & AMD—the Conroe and AM2 processors respectively.